



Procedure #5

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CLAYTON BOILER TRIP CONDITION

RE-START AFTER HIGH PRESSURE/TEMPERATURE TRIP PROCEDURE

This procedure is also referred to as a "wet start up". (This process is required often, and is typically completed weekly.)

Complete a thorough check over of the boiler before re-setting any trip switch.

Note: Use extreme caution. Before pressing the Stop button, note the trip condition display message on both the Clayton OIU and Siemens PLC; either should provide details as to why there is a trip occurring.

ASSUMPTIONS:

- The operator is familiar with the steps of this procedure before commencing.
- The steam system is still pressurized and ready to receive steam from the Clayton Boiler

Assess the Reason for the Trip			
Step	Operation	Initial when Complete	Date and Time when Complete
1	Ensure the steam system is still pressurized and ready to receive steam from the Clayton Boiler.		
2	Note the trip condition display message on both the Clayton OIU and Siemens PLC before pressing the Red-Stop-Button ; either should provide details as to why there is a trip occurring.		
3	Press the Red-Stop-Button to silence alarm.		
4	Assess the reason for trip: either high pressure or high coil temperature.		
5	Assess the reason for trip: either high pressure or high coil temperature.		

Boiler Tripped on High Coil Temperature			
Step	Operation	Initial when Complete	Date and Time when Complete
6	Place the Clayton OIU into Manual and into Fill mode.		
7	Partially open the valve from the bottom of the separator to direct water to the blow down tank to allow water to flow through the boiler for about 10 minutes while monitoring the separator temperature		
Note: The boiler FD fan can be manually turned on to assist cooling the boiler coil. The switch for the fan is located inside the boiler control cabinet.			
8	Verify the temperature on the separator has dropped below reset limits.		
9	Reset the Coil High Temperature Trip Switch (ATS) X2 		
10	Assess the reason for the cause of the high coil temperature.		
11	If safe, restart the boiler, starting from step 14 of the Clayton Boiler Startup Procedure.		
Boiler Tripped on High Steam Pressure			

12	Allow pressure the steam pressure to drop to below the high-steam-pressure switch setting.	Initial when Complete	Date and Time when Complete
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Step	Operation		
13	Reset the Clayton Boiler high steam pressure trip switch. 		
14	If safe, restart the boiler, starting from step 14 of the Clayton Boiler Startup Procedure. Leave the main stop valve open if the plant is still hot		
“Air Pressure Lock-Out” Alarm on Flame Monitor			
15	Press red-stop-button, then press ESC, at “Do you want to clear this alarm?” quickly press enter button on the Siemens panel to re-set this alarm.		
High Separator Level Trip			
16	Silence the alarm by pressing the red-stop-button. Reset the boiler and start from step 14 of the Clayton Boiler Startup Procedure.		
17	Make an appropriate log book entry.		

Note: Troubleshooting information can be found on page 5-1 of the Clayton Industries Steam Generator Instruction Manual. The following additional alarm conditions also require operator response through the Siemens PLC and the Clayton OIU:

1. High pressure
2. High temperature
3. Low gas pressure
4. High gas pressure
5. Pilot flame failure
6. Main burner failure
7. FD fan failure
8. Cooling fan failure
9. Fuel Gas – has own reset switch

End of Procedure